

Guangrui Li

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EDUCATION

University of Technology Sydney

Ph.D. in Computer Science

Sydney, Australia

July 2019 – May 2024

Chongqing University

B.E. in Software Engineering

Chongqing, China

Sep. 2014 – June 2018

RESEARCH INTEREST

Keywords: Transfer Learning, Multi-Modal Learning, Open-world Learning.

- **Vision-Language & Foundation Models::** Investigating the reasoning capabilities of Multi-modal Large Language Models (MLLMs) and developing robust fine-tuning strategies for Vision Foundation Models.
- **AI Robustness & Generalization::** Focusing on enhancing model reliability through techniques in Domain Adaptation , Out-of-Distribution (OOD) generalization , and Federated Learning.
- **3D Vision & Scene Understanding.** Applying deep learning to challenges in point cloud segmentation , animal pose estimation , and video scene parsing.

PROFESSIONAL EXPERIENCE

Amazon Search, Rufus

Applied Scientist II

Palo Alto, CA

06/02/2025 – Now

- Architected a multi-agent routing system from the ground up, achieving over 95% accuracy in directing user queries to specialized downstream agents.
- Reduced Rufus agent latency by deploying a 5x smaller model while preserving core helpfulness metrics, enabled by strategic workflow decomposition.

AWS AI Rekognition

Research Intern on Vision Foundation Models

Seattle, WA

11/06/2023 – 02/09/2024

- Worked on enhancing the content moderation services in AWS.
- Focused on improving the robustness and generalizabilities during the fine-tuning of vision foundation models (CLIP, DINOv2, etc), striking the balance between on in-distribution (ID) recognition and robustness to out-of-distribution (OOD) samples.
- Based on the internship project, first-authored a paper that latterly was accepted by ECCV 2024.

Sony AI Research

Research Intern on Vision Foundation Models

Remote

08/21/2023 – 10/30/2023

- Studied Federated Knowledge Distillation for visual foundation models, i.e., gathering and distilling detection knowledge from multiple distributed detectors.
- Collaborated with a team of researchers to develop and integrate a all-in-one vision foundation models.
- Implemented novel techniques for effective class aggregation in federated knowledge distillation.

Baidu Research

Research Intern on Multi-Modal Learning

Beijing, China

Sep 2021 – Aug 2022

- Developed novel methodologies for open-vocabulary detection to identify and localize novel/unseen objects based on language prompts.
- Achieved state-of-the-art results on COCO and LVIS benchmarks, first-authored a research paper.
- The developed approach has been examined and deployed in Baidu Services.

PUBLICATIONS

1. RAFT: Reasoning-Aware Fine-Tuning for MLLMs
Submitted to Neural Information Processing Systems (**NeurIPS**), 2025. [In submission]
Jinhao Li, Guangrui Li, Riheng Zhu, Zhenbang Sun, Feng Liu.
2. Robustness Preserving Fine-tuning using Neuron Importance
European Conference on Computer Vision (**ECCV**), 2024.
Guangrui Li, Rahul Duggal, Aaditya Singh, Kaustav Kundu, Bing Shuai, Jonathan Wu.
3. Construct to Associate: Cooperative Context Learning for Domain Adaptive Point Cloud Segmentation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
Guangrui Li.
4. Decouple to Contrast: Orthogonalized Ambiguity Reduction for Open-Vocabulary Object Detection
Guangrui Li, Yifan Sun, Zongxin Yang, Errui Ding, Yi Yang
5. Adversarially Masking Synthetic to Mimic Real: Adaptive Noise Injection for Point Cloud Segmentation Adaptation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.
Guangrui Li, Guoliang Kang, Xiaohan Wang, Yunchao Wei, and Yi Yang
6. Decompose to Generalize: Species-Generalized Animal Pose Estimation
International Conference on Learning Representations (**ICLR**), 2023.
Guangrui Li, Yifan Sun, Zongxin Yang, and Yi Yang
7. VSPW: A Large-scale Dataset for Video Scene Parsing in the Wild
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
Jiaxu Miao, Yunchao Wei, Yu Wu, Chen Liang, Guangrui Li, Yi Yang
8. Domain Consensus Clustering for Universal Domain Adaptation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
Guangrui Li, Guoliang Kang, Yi Zhu, Yunchao Wei, and Yi Yang
9. Content-Consistent Matching for Domain Adaptive Semantic Segmentation
European Conference on Computer Vision (**ECCV**), 2020.
Guangrui Li, Guoliang Kang, Wu Liu, Yunchao Wei, and Yi Yang

ACADEMIC SERVICES AND AWARDS

Google Travel Grant, 2023
ICLR financial assistance award, 2023
Semi-finalist in Valeo International Challenge [24/1376], 2017
Reviewer of IEEE TPAMI, TKDE, TIP, and TNNLS.
Programme Committee of CVPR, ECCV, ICCV, ICLR, ICML, and NeurIPS.

SKILLS

Programming: Python (Numpy, Pytorch, Caffe), C/C++, Shell, $\text{\LaTeX}$.
Languages: Native in Chinese (Mandarin), Fluent in English.